

JEE MAIN REVISION MASTER: PERIODIC TABLE AND PERIODICITY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 16 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Periodic Trends: Radii, Ionization Enthalpy & Electronegativity	16 PYQs	Periodic Trends

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Periodic Trends: Radii, Ionization Enthalpy & Electronegativity	$\text{Isoelectronic Radii: } \text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Periodic Trends: Radii, Ionization Enthalpy & Electronegativity (Very High Yield — 16 PYQs)

Core Principle: Periodic trends across a period (left to right): Atomic radius decreases, ionization enthalpy increases, electronegativity increases, electron gain enthalpy generally becomes more negative. Down a group: Atomic radius increases, ionization enthalpy decreases. Isoelectronic ions: size decreases with increasing positive nuclear charge (Z). Anomalies: $IE_1(\text{Be}) > IE_1(\text{B})$ (fully filled $2s^2$), $IE_1(\text{N}) > IE_1(\text{O})$ (half-filled $2p^3$).

■ **Caution / JEE Pitfall:** Electron gain enthalpy of Chlorine is MORE negative than Fluorine (-349 kJ/mol vs -328 kJ/mol) due to high electron-electron repulsion in small 2p subshell of F!

■ **Speed Trick:** Second ionization enthalpy of alkali metals is extraordinarily high because removing a second electron breaks a stable noble gas core.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Match List-I with List-II: List-I (Element-Atomic number) List-II (Position in periodic table) (a) Ra – 88 (i) 4th period, 13th gr..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Given below are the atomic masses of the elements: $\begin{matrix} \text{Element:} \\ \text{Li} & \text{Na} & \text{Cl} & \text{K} & \text{Ca} & \text{Br} & \text{Sr} & \text{I} & \text{Ba} \end{matrix}$..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "In which one of the following pairs, both the elements does not have $(n-1)d^{10}ns^2$ configuration in its elementary state?" [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Match List-I with List-II: List-I (Element-Atomic number) List-II (Position in periodic table) (a) Ra – 88 (i) 4th period, 13th group (b) Ga – 31 (ii) 6th period, 6th group (c) W – 74 (iii) 5th period, 10th group (d) Pd – 46 (iv) 7th period, 2nd group Choose the correct answer from the options given below.

- (A) a-iv, b-i, c-ii, d-iii
- (B) a-i, b-ii, c-iii, d-iv
- (C) a-iv, b-ii, c-iii, d-i
- (D) a-iii, b-iv, c-i, d-ii

Correct Answer: None

Solution Analysis: For finding the position of an element in the periodic table, we use its electronic configuration. (a) Ra, atomic number 88 Ra belongs to alkaline earth metals. Its outer configuration is: $\text{Ra}:[\text{Rn}]7s^2$ Highest shell number is 7, so it is in the 7th period. It has 2 valence electrons in the s-subshell, so it is in the 2nd group. So, Ra matches with (iv). (b) Ga, atomic number 31 Electronic configuration: $\text{Ga}:[\text{Ar}]3d^{10}4s^24p^1$ Highest shell number is 4, so it is in the 4th period. It has configuration ns^2np^1 , so it belongs to group \$